Michael Kim

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Education

University of Florida, Gainesville, FL — *Expected: August 2026*

Ph.D. Statistics GPA 3.73/4.00

Research Topics: Bayesian Models | Missing Data | Markov Chain Monte Carlo (MCMC)

University of North Carolina at Chapel Hill, Chapel Hill, NC — August 2013 – May 2017

B.S. Statistics **B.A.** Mathematics GPA 3.629/4.000

Research Experience

Graduate Research Assistant — August 2021 – September 2025
University of Florida — Gainesville, FL

Bayesian Structured Correlation Modeling for Biomarker Selection in Longitudinal DMD Data

- Designed a novel Metropolis–Hastings algorithm that enforces positive definiteness of structured correlation matrices more efficiently than standard rejection-based methods by tightening the support of the proposal distribution to be narrower than the standard $(-1, 1)$ range.
- Developed a structured covariance matrix to address missingness in longitudinal Duchenne muscular dystrophy (DMD) data, specifying inter-muscle and temporal correlation parameters.
- Identified the most responsive biomarkers (annualized changes in fat fraction of muscles) to disease progression of DMD by constructing biomarker composites that maximize the standardized response mean at different ambulatory stages.
- Validated the robustness of the proposed MCMC approach via large-scale simulations over thousands of datasets, demonstrating stable convergence, high positive-definiteness rates, accurate credible interval coverage, low bias, and low mean squared error.

Graduate Research Assistant — *September 2025 – Present*
University of Florida — Gainesville, FL

Modeling Optimal Biomarker Composites as a Continuous Function of 6MWD

- Implemented a multivariate nonlinear mixed-effects (NLME) model in Stan to simultaneously model six concurrent trajectories (five leg muscles and six-minute walking distance (6MWD)), utilizing observed-data likelihoods to handle missingness and 10*m* floor-censoring.
- Developed a grid-based algorithm to construct optimal biomarker composites (one-year MRI-T2 changes of leg muscles) as a continuous function of 6MWD, generalizing on existing discrete-stage frameworks for quantifying disease progression.
- Conducted Bayesian model selection to determine the random-effects structure, evaluating predictive performance via WAIC and parameter identifiability through posterior shrinkage and relative standard errors.
- Establishing asymptotic properties, including posterior consistency and asymptotic normality, for model parameters and optimal composite biomarker weights.

Publications

Kim M, Daniels MJ, Rooney WD, Willcocks RJ, Walter GA, Vandenborne KH. A new algorithm for sampling parameters in a structured correlation matrix with application to estimating optimal combinations of muscles to quantify progression in Duchenne muscular dystrophy. *Statistics in Medicine*, 44 (2025) 20-22.

Carmichael I, Wudel J, **Kim M**, Jushchuk J. Examining the Evolution of Legal Precedent through Citation Network Analysis. *North Carolina Law Review*, 96 (2017) 227-269.

Kim M, Daniels MJ, Kim Sarah, Yoon DY, Rooney WD, Willcocks RJ, Walter GA, Vandenborne KH. A novel approach to optimize composite biomarkers for continuous measures of clinical function: Application to muscle imaging biomarkers in Duchenne muscular dystrophy. (2026). In preparation.

Skills

Languages: R | Stan | SAS | Python | LaTeX
Statistics: Hierarchical Bayesian Modeling | Missing Data Analysis | MCMC Algorithms | Joint Modeling

Coursework

MCMC | Probability Theory | Generalized Linear Models | Advanced Regression | Design of Experiments | Time Series | Optimization | Machine Learning

Awards

Grinter Fellowship — *August 2017 - Spring 2020*